Regression Discontinuity Designs

PS200C, Spring 2026

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A motivating case: Brazilian electronic voting

Brazil introduced electronic voting machines in 1998, replacing paper ballots.

Roll-out was **not** based on need, ideology, or party — it was based on a **population rule**:

- Municipalities with $>$ 40,500 registered voters $\rightarrow$ e-voting in 1998
- Municipalities at or below the threshold $\rightarrow$ paper ballots

Hidalgo (2012) asks: did e-voting change *who* got represented?

- Paper ballots required reading and writing candidate numbers; many voters spoiled their ballots.
- E-voting effectively enfranchised millions of less-educated voters.

What's the causal effect of e-voting on the share of valid votes (and downstream, on left-leaning vote share, social spending, etc.)?

Why we can't just compare

Naive: compare e-voting municipalities to paper-ballot ones.

But the threshold cut the country into two very unequal halves:

- Above threshold: bigger, richer, more urban, more literate, more developed.
- Below threshold: small, rural, poorer.

A selection on observables strategy would require trying to catch all such confounding differences. Plus, we know there is not common support (on population).

The RD trick: Give up on the ATE; just compare *near the cutoff* where a small variation in municipality population would change its treatment status.

Result

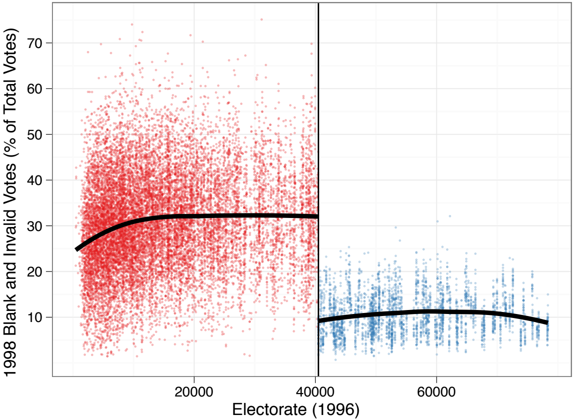


Figure 6: The effect of electronic voting on the percent of null and blank votes. Each dot is a polling station. Polling stations to the left of the vertical black line used paper ballots and polling stations to the right used electronic voting. The black horizontal line is the conditional mean of the outcome estimated with a loess regression.

Generalizing the idea

A binary treatment D_i that is **determined** by whether a continuous predictor X_i crosses a fixed threshold c:

$$D_i = \mathbb{1}\{X_i > c\} \quad \Longleftrightarrow \quad D_i = \begin{cases} 1 & \text{if } X_i > c \\ 0 & \text{if } X_i \leq c \end{cases}$$

This is common in rule-based settings: administrative programs, eligibility cutoffs, electoral rules, age cutoffs...

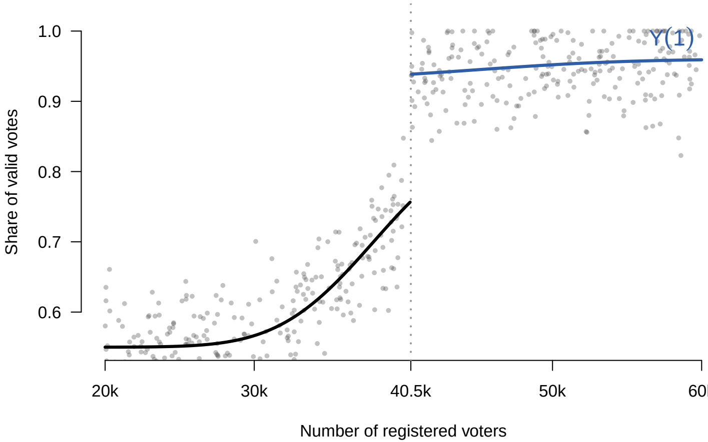
X_i: the forcing or running variable; c: the cutoff or threshold.

X_i may be (and usually is) correlated with the potential outcomes. This isn't an experiment.

Later: *fuzzy* RD, where $X_i > c$ **shifts the probability** of D rather than determining it.

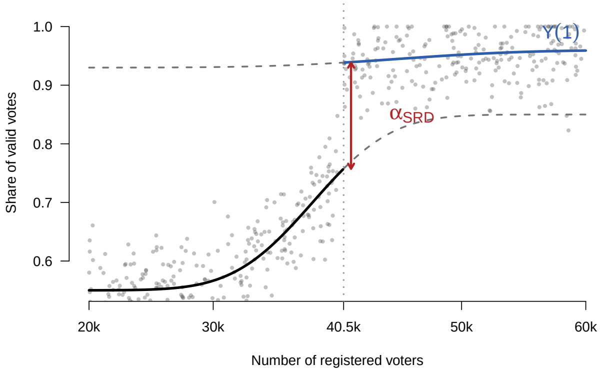
Continuity of potential outcomes

Each municipality has two potential outcomes: $Y(0)$ if paper, $Y(1)$ if e-voting, but we see only one or the other.



Continuity of potential outcomes

Really, think of this in terms of two objects: $\mathbb{E}[Y(0) \mid X]$ and $\mathbb{E}[Y(1) \mid X]$:



The effect α_{SRD} is the vertical gap between the tracks *at the cutoff*.

The core assumption is **continuity**: both tracks vary *smoothly* with X at c.

Sharp RD: identification

Identification Assumption (Continuity of conditional expectations)

$\mathbb{E}[Y_{1i} \mid X_i]$ and $\mathbb{E}[Y_{0i} \mid X_i]$ are continuous in X_i at $X_i = c$.

Plain-language: Both potential outcomes CEFs move *smoothly* across the running variable, at least at the cutoff.

Identification Result

The treatment effect at the threshold is identified as:

$$\begin{aligned} \alpha_{SRD} &= \mathbb{E}[Y_{1i} - Y_{0i} \mid X_i = c] \\ &= \lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x] \end{aligned}$$

*Without further assumptions, α_{SRD} is **only valid at the threshold**.*

Comments on identification

- We are only saying that $Y(0)$ and $Y(1)$ don't suddenly *jump* at $X = c$.
    - We are *not* saying $Y(0)$ and $Y(1)$ are unrelated to X,
    - We are *not* saying $Y(0)$ and $Y(1)$ are flat or have the same shape.
- The main thing that would violate this: another treatment kicking in at the same cutoff.
    - If something else changes at $X = c$ that affects Y, then even $Y(0)$ (and $Y(1)$) jump – i.e. a discontinuity.
    - Plainly, RD would mis-attribute the effect of another treatment to the one you are thinking about.

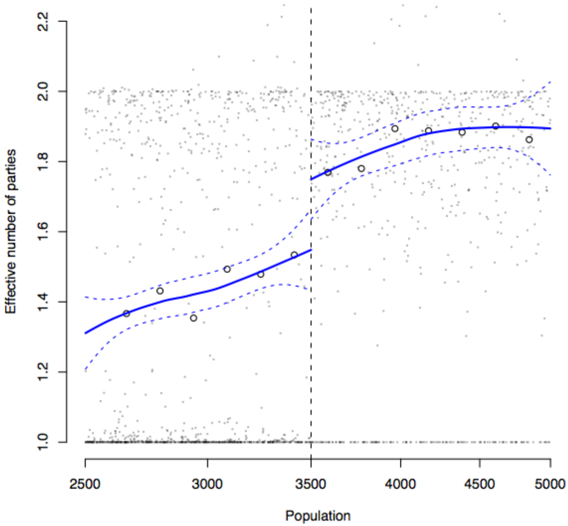
This is *the main* threat to keep in mind. We'll return to it.

Duverger's Law (Eggers, 2010)

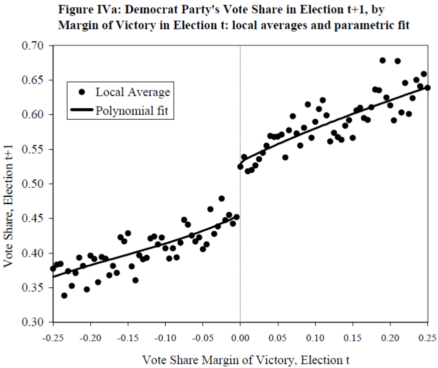
- **Duverger (1972)**: "a majority vote on one ballot is conducive to a two-party system; proportional representation is conducive to a multiparty system."
- Prediction: more parties under PR than under FPTP.
- French municipal elections use a population rule:
  - $<$ 3,500 residents $\rightarrow$ first-past-the-post
  - $\geq$ 3,500 $\rightarrow$ proportional representation

X_i = population, c = 3,500, D_i = PR rule, Y_i = number of parties.

Duverger's Law: the jump



Incumbency advantage (Lee, 2008)



- X_i = Dem party vote margin in last election,
- $c = 0$, D_i = won the seat,
- Y_i = Dem party vote share next election.

A few others

- Effect of class size on student achievement (Angrist & Lavy 1999, Maimonides' rule)
- Effect of access to credit (loan offer determined by credit score threshold)
- Effect of mayoral wages on performance (Ferraz & Finan 2011)
- Effect of winning office on politicians' wealth (Eggers & Hainmueller 2009)
- Effect of school district boundaries on home values (Black 1999)
- "Close election" RD has become a workhorse for electoral research: Lee, Moretti & Butler 2004; Pettersson-Lidbom 2008; Eggers & Hainmueller 2009; many more.

Warning: "close election" designs with treatment other than *winning* (party, gender, education) measure the effect of *electing somebody with that characteristic* – think districts, not individuals.

How to actually estimate

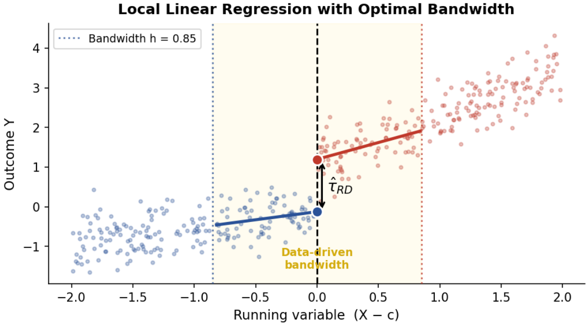
Historically, or when casually exploring the data, you might try some simple polynomial models:

- Choose a window of the data somewhat near the cutoff to look at.
- Transform your X to be zero at $X = c$, i.e. $\tilde{X} = X - c$.
- Use a linear, quadratic, or cubic model plus a shift at $X = c$; better to allow different slopes on each side
- E.g., $Y \sim \tilde{X} + \tilde{X}^2 + D + \tilde{X}D + \tilde{X}^2 D$

But, there are important underlying estimation choices to make here that are better addressed by certain estimators. We will discuss two.

Both are local: they fit $\mathbb{E}[Y \mid X]$ on each side of c using observations *near* c, and read off the gap at c.

Option 1: `rdrobust` (Calonico, Cattaneo, Titiunik 2014)



- Local linear regression on each side, triangular kernel
- $\hat{\alpha}$ = gap in fitted values at c
- Automated, principled bandwidth selection
- Bias-corrected, robust CIs

```
rdrobust(y, x, c = 0)
```

Option 2: Gaussian process (`gpss::gp_rdd`)

Two risks of any local-polynomial fit:

- Polynomials happily extrapolate near the edge — where they have least data and most leverage.
- Uncertainty should *grow* as we approach c because we have less data around those points.

The Gaussian process:

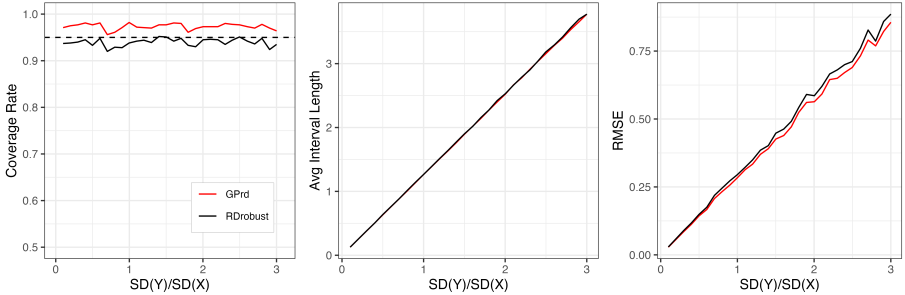
- Fits models in a flexible space of smooth functions
- "Carries" an honest range of plausible curve fits through the data $\rightarrow$ increased uncertainty towards and past edge of the data.
- Regularizes model on each side to avoid misreading a few noisy points as a radically different curve.
- Fully automated (Cho, Kim & Hazlett 2025): `gpss::gp_rdd(x, y, cut)`

In practice

- Large samples near the cutoff: `rdrobust` and GP agree.
- Small samples: GP gives better estimates, better coverage, fewer extreme results.
- Very small samples: `rdrobust` sometimes chooses very wide bandwidth or generates extreme results or NA.

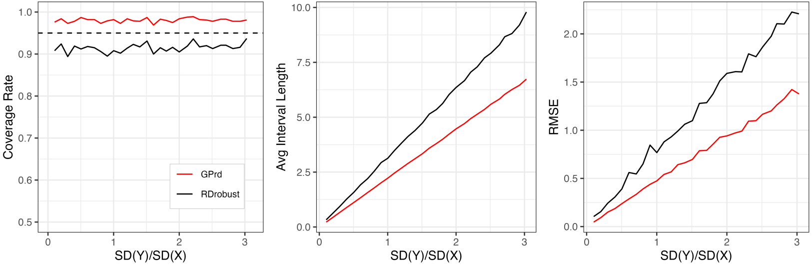
Random Z and Y, large sample

$\tau \in \{0, 3\}$, $N = 500$, `rdrobust` at defaults vs `gp_rdd` at defaults.



Random Z and Y, small sample

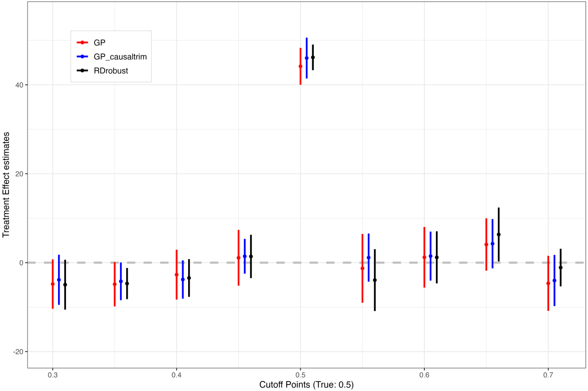
Same DGP, $N = 100$.



Improved RMSE is the big story — GP regularizes on each side, where polynomials can go a little crazy.

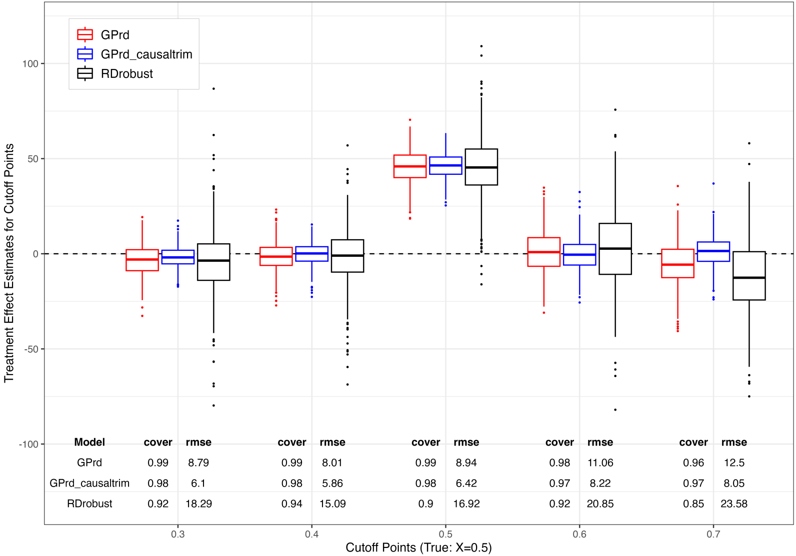
Lee close elections, full sample ($N = 13{,}577$)

Placebo cutoffs across Lee's close-election data. With this much data, methods barely diverge.



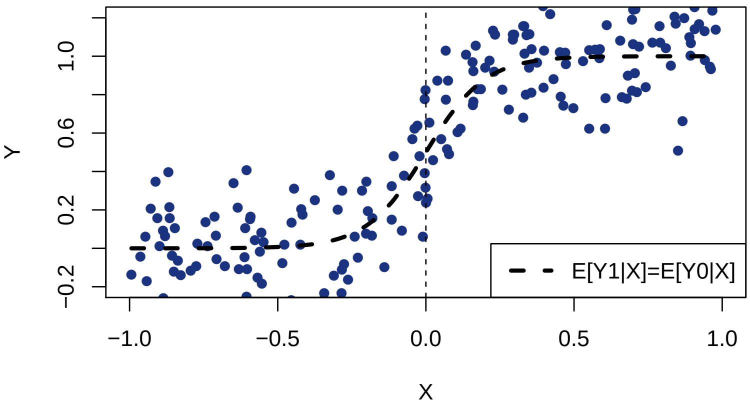
Lee close elections, small subsamples ($N = 200$)

Repeat sampling with $N = 200$ per segment. Now method choice matters.

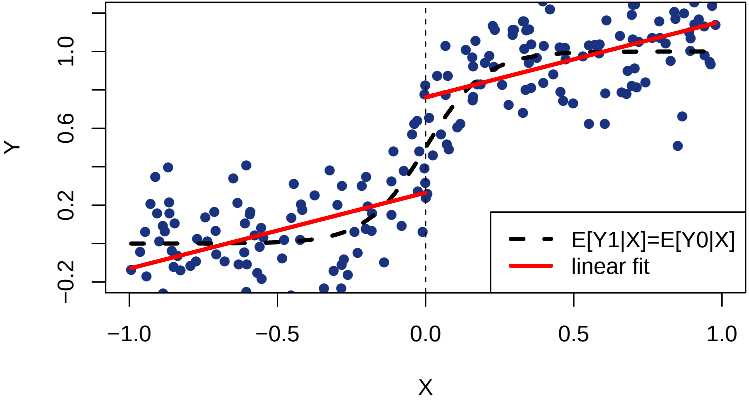


Model	cover	rmse	cover	rmse	cover	rmse	cover	rmse	cover	rmse
GPrd	0.99	8.79	0.99	8.01	0.99	8.94	0.98	11.06	0.96	12.5
GPrd_causaltrim	0.98	6.1	0.98	5.86	0.98	6.42	0.97	8.22	0.97	8.05
RDrobust	0.92	18.29	0.94	15.09	0.9	16.92	0.92	20.85	0.85	23.58

The risk of misspecification

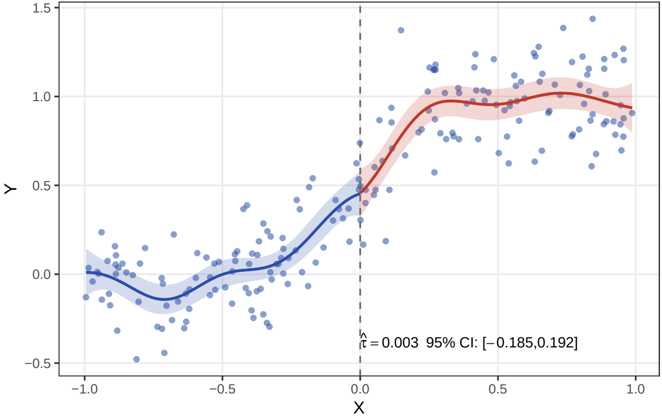


The risk of misspecification



Same example, GP estimator

Should be no problem at all with a suitably flexible estimator.

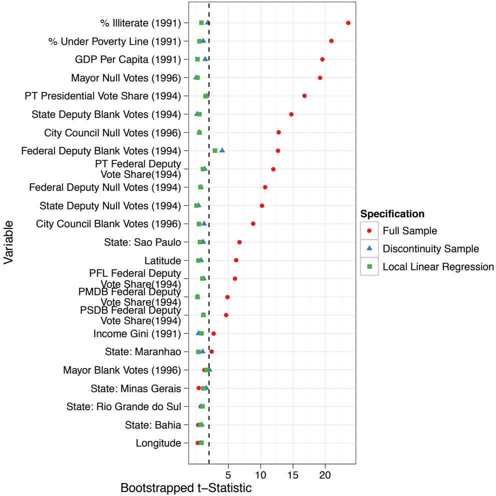


`gpss::gp_rdd(x, y, cut=0).`

Falsification checks: the menu

1. **Specification sensitivity**: are results stable across estimators / bandwidths? If not, explain.
2. **Balance**: do covariates Z jump at the threshold?
3. **Placebo outcomes**: does an outcome that *shouldn't* respond to treatment also jump?
4. **Placebo thresholds**: do you find jumps at $c^\star \neq c$?
5. **Sorting**: do units cluster on one side of the cutoff (manipulation)?

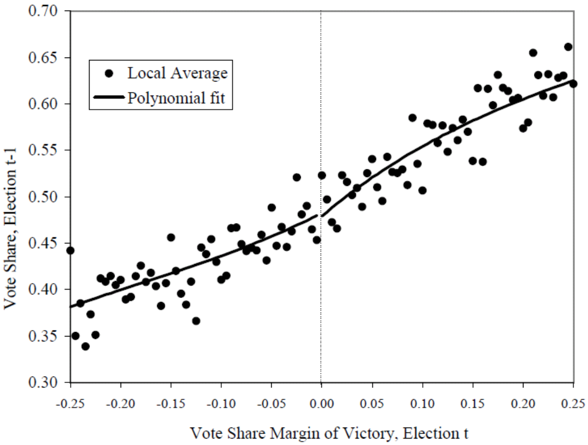
Balance test, back to Hidalgo



Run the same RD on a covariate Z as the “outcome” — hope for no jump.

Placebo outcome (Lee 2008)

Most convincing falsification: a covariate (or pre-treatment outcome) plausibly related to potential outcomes but *not* affected by treatment.



Placebo thresholds and sorting

Placebo thresholds: pretend the cutoff is $c^\star \neq c$. Run the RD.

- Detecting a jump at $c^\star$ demands an explanation.
- ...Are there other treatments at $c^\star$? Some reason for non-smoothness?

Sorting: can units *self-select* across the cutoff?

- Fine control over X (gaming a test score, withholding income).
- E.g. Administrators choosing thresholds strategically.
- Density test (McCrary 2008): bulge in X on one side of c?
- Not always a problem: sorting would have to be very precise, and widespread, to substantially bias the result.

The main worry: bundled treatments

- Sometimes *multiple things* change at the same threshold.
- This is bad:
  - If a *different* treatment affects Y at the same cutoff then your outcome Y will be hit regardless of the treatment you are asking about.
  - That is, your $Y(0)$ and $Y(1)$ will jump at the cutoff – violation of the continuity assumption.
  - Essentially a confounder right at the cutoff.
- In other words, the RD does not tell you *what* treatment effect you are seeing.
- Always ask: *what else changes at c?*. This is where careful investigatory work and domain expertise pay off.

Fuzzy RD

- The threshold doesn't *determine* treatment, but it **shifts the probability** of treatment.
- E.g.: scoring above a threshold raises the chance of getting a scholarship from 20% to 70%, but doesn't guarantee it.
- This should remind you of IV.
    - Crossing c is the instrument.
    - It changes treatment probability and (we hope) only affects Y through treatment.

Doing fuzzy RD

The recipe (just like IV at the cutoff):

1. **Reduced form**: estimate the jump in Y at the cutoff (the sharp RD on Y).
2. **First stage**: estimate the jump in D at the cutoff (the sharp RD on D).
3. Divide.

Toy numbers: crossing c raises income by \$5,000 *and* raises scholarship probability from 20% to 70%.

Compute $\hat{\alpha}_{\text{LATE}} = 5000/0.5 = \$10{,}000$.

Interpretation and credibility:

- The estimand is **doubly local**: at the cutoff, and only for compliers.
- Same threat as before: no *other* treatments switching at c.

`rdrobust` can do this for you (`fuzzy = D`).

Internal vs. external validity

Internal validity:

- Continuity of POs is often defensible, making RD one of the happiest observational approaches.
- Main threat: **bundled treatments** at the same cutoff.

External validity:

- You learn the average effect for units with X_i *near* c.
- Fuzzy RD: also restricted to compliers near c.
- No information about effects elsewhere on the X distribution. Don't try.
- Often useful in combination with other approaches such as SOO that aim for broader target but have less credible assumptions.

Takeaways

- RD turns an administrative rule into something close to an experiment — locally.
- Identification: **continuity**. Plain English: no other (impactful) treatments at same cutoff.
- Estimation: use an automated tool (`rdrobust`, `gp_rdd`)
- Falsification battery: balance/ placebo outcomes, placebo thresholds, sometimes sorting
- But most important is investigatory work to look for other treatments at same cutoff.